



CREDIT CARD
ANALYTICS

FINANCIAL ANALYSIS

Unlocking Insights. Driving Performance. Managing Risk.



Customer
Insights



Transaction
Analysis



Revenue
Performance



Risk
Management



Prepared By
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Analysis Period
JAN 2023 – DEC 2023



Tools Used
POWER BI | DAX

“ A comprehensive view of customer behavior,
financial performance & credit risk. ”





PROJECT OVERVIEW & BUSINESS OBJECTIVES



PROJECT OVERVIEW

This project aims to analyze credit card customer behavior, transaction patterns, credit utilization, revenue performance and repayment risk using Power BI and DAX.



BUSINESS PROBLEM

Banks need deeper insights into how customers use their credit cards, their spending and repayment behavior, and the associated risk, in order to improve customer retention, reduce delinquencies and enhance overall financial performance.



01

TRANSACTION ANALYSIS

Analyze transaction trends, running totals and growth patterns to understand performance.



02

CUSTOMER BEHAVIOR

Understand credit utilization, spending behavior and customer segments.



03

REVENUE ANALYSIS

Evaluate transaction value, interest income and acquisition cost to measure profitability.



04

RISK MANAGEMENT

Identify delinquent, high-utilization and high-risk customers to minimize credit risk.



05

CUSTOMER RETENTION

Detect inactive/churned customers and create insights for retention strategies.



GOAL:

Convert raw credit card data into actionable insights that help improve customer experience, maximize revenue and strengthen risk management.





DATASET OVERVIEW & DATA MODEL

DATASET OVERVIEW



10,108

Total Customers



2023

Analysis Period



₹44.52M

Total Transaction Amount



₹87.29M

Total Credit Limit



₹11.75M

Total Revolving Balance



₹7.84M

Interest Earned



₹0.97M

Customer
Acquisition Cost

DATASET TABLES



Customer

Contains customer demographic and profile information.



Credit Card

Contains credit card details and account information.



Customer Additional

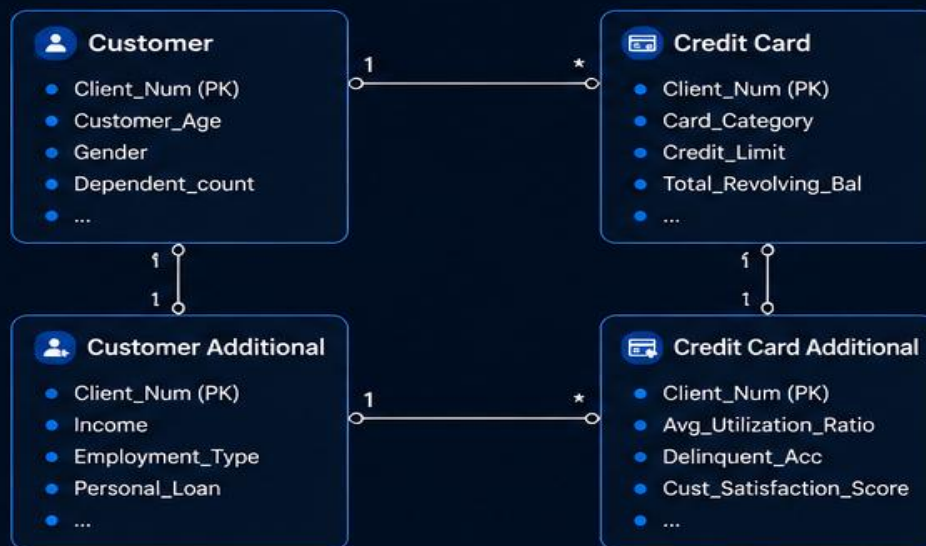
Contains additional customer details such as income, employment, loan status etc.



Credit Card Additional

Contains additional credit card related details like utilization, satisfaction score, delinquency etc.

DATA MODEL



RELATIONSHIP LEGEND

1 One
* Many



The data model integrates customer, credit card and additional information to provide a unified view for in-depth financial analysis.



Integrated data from multiple sources to build a robust model for analyzing customer behavior, transaction performance, revenue, and credit risk.





Q1. RUNNING TOTAL OF CREDIT CARD TRANSACTIONS

SLIDE
04

Q1 Running Total of Credit Card Transactions

Year	Month	Total_Transaction_Amount (₹)	Running_Total (₹)
2023	January	43,22,186	43,22,186
2023	February	35,39,575	78,61,761
2023	March	33,88,827	1,12,50,588
2023	April	41,74,728	1,54,25,316
2023	May	34,26,913	1,88,52,229
2023	June	35,33,660	2,23,85,889
2023	July	45,46,958	2,69,32,847
2023	August	34,49,868	3,03,82,715
2023	September	34,52,874	3,38,35,589
Total		4,55,33,021	4,55,33,021



Total Transaction Amount (2023)

₹4,55,33,021



ABOUT THIS METRIC

The Running Total represents the cumulative value of credit card transactions over time, helping to understand the overall growth in transaction volume throughout the year.



DAX FORMULA USED

```
Running_Total =  
CALCULATE(  
    SUM('Credit Card Additional'[Total_Trans_Amt]),  
    FILTER(  
        ALLSELECTED('Calendar'[Date]),  
        'Calendar'[Date] <= MAX('Calendar'[Date])  
    )  
)
```



KEY INSIGHTS

- Transactions show steady growth across the year.
- Highest single month transactions recorded in July (₹45.47M).
- Cumulative transactions reached ₹4.55 Cr by September 2023.



TAKEAWAY

Running total analysis helps in tracking cumulative performance and identifying periods of strong or weak transaction activity.





Q2. 4-WEEK MOVING AVERAGE OF CREDIT LIMIT

Q2 4-Week Moving Average of Credit Limit

Year	WeekNumber	Total_CreditLimit (₹)	Moving_Average_4_Weeks (₹)
2023	1	17,04,635.70	17,04,636
2023	2	16,00,959.70	16,52,798
2023	3	15,26,415.10	16,10,670
2023	4	16,47,543.20	16,19,888
2023	5	17,98,645.00	16,43,391
2023	6	17,10,629.70	16,70,808
2023	7	19,76,267.00	17,83,271
2023	8	17,33,164.80	18,04,677
2023	9	15,07,517.70	17,31,895
2023	10	14,41,620.20	16,64,642



ABOUT THIS METRIC

The 4-week Moving Average smooths short-term fluctuations in Credit Limit and helps identify underlying trends over time.



DAX FORMULA USED

```
Moving_Average_4_Weeks =  
var time_period = FILTER(ALL(Calendar),  
    Calendar[WeekNumber] > MAX(Calendar[WeekNumber]) - 4  
    && Calendar[WeekNumber] <= MAX(Calendar[WeekNumber]))  
var creditLimit_time_period =  
    CALCULATE([Total_CreditLimit],time_period)  
var distinct_week =  
    CALCULATE(DISTINCTCOUNT(Calendar[WeekNumber]),time_period)  
return DIVIDE(creditLimit_time_period, distinct_week)
```



KEY INSIGHTS

- 4-week moving average provides a clearer view of credit limit trend by reducing week-to-week volatility.
- Useful for decision making on credit limit strategy and customer segmentation.



TAKEAWAY

The 4-week moving average helps track the stable trend in credit limits and supports better planning for risk management and customer growth.





Q3. MOM% GROWTH & WOW% GROWTH ON TRANSACTION AMOUNT



OVERVIEW

Month over Month (MoM) growth compares each month's transaction amount with the previous month.

Week over Week (WoW) growth compares each week's transaction amount with the previous week.



DAX FORMULAS USED

MONTH OVER MONTH GROWTH

```
MOM_GROWTH =  
VAR prev_month =  
    CALCULATE([Total_Transaction_Amount],  
        DATEADD(Calendar[Date], -1, MONTH))  
RETURN DIVIDE(  
    [Total_Transaction_Amount] - prev_month,  
    prev_month, 0)
```

WEEK OVER WEEK GROWTH

```
WOW_GROWTH =  
VAR prev_week =  
    CALCULATE([Total_Transaction_Amount],  
        DATEADD(Calendar[Date], -1, WEEK))  
RETURN DIVIDE(  
    [Total_Transaction_Amount] - prev_week,  
    prev_week, 0)
```



MoM GROWTH (LATEST MONTH)

17.32%

October 2023 vs September 2023



WoW GROWTH (LATEST WEEK)

-10.07%

Week 10 vs Week 9

MONTH OVER MONTH GROWTH

Year	Month	Total_Transaction_Amount (₹)	MOM_GROWTH
2023	January	43,22,186	0.00%
2023	February	35,39,575	-18.11%
2023	March	33,88,827	-4.26%
2023	April	41,74,728	23.19%
2023	May	34,26,913	-17.91%
2023	June	35,33,660	3.11%
2023	July	45,46,958	28.68%
2023	August	34,49,868	-24.13%
2023	September	34,52,874	0.09%
2023	October	40,50,909	17.32%

WEEK OVER WEEK GROWTH

Year	WeekNumber	Total_Transaction_Amount (₹)	WOW_GROWTH
2023	1	8,35,767	0.00%
2023	2	8,44,739	1.07%
2023	3	9,23,367	9.31%
2023	4	8,69,235	-5.86%
2023	5	8,49,078	-2.32%
2023	6	8,98,867	5.86%
2023	7	8,90,756	-0.90%
2023	8	8,68,091	-2.54%
2023	9	8,81,861	1.59%
2023	10	7,93,080	-10.07%



KEY INSIGHTS



Highest MoM Growth:
July 2023 (28.68%)



Lowest MoM Growth:
August 2023 (-24.13%)



Highest WoW Growth:
Week 3 (9.31%)



Lowest WoW Growth:
Week 10 (-10.07%)



Q4. CUSTOMER ACQUISITION COST (CAC) AS A RATIO OF TRANSACTION AMOUNT



OVERVIEW

This metric shows the Customer Acquisition Cost (CAC) as a percentage of the Total Transaction Amount generated by each client.

OVERALL SUMMARY

TOTAL CAC



₹9,72,148

Total Customer
Acquisition Cost

TOTAL TRANSACTION AMOUNT



₹4,55,33,021

Total Transaction
Amount

CAC / TRANSACTION RATIO (OVERALL)



2.14%

CAC as a % of
Transaction Amount



INSIGHT

On average, the bank spends **2.14%** of the transaction value to acquire customers.



DAX FORMULA USED

```
CAC_Ratio =  
DIVIDE(  
    [Customer_Acquisition_Cost],  
    [Total_Transaction_Amount],  
    0  
)
```

$$\text{CAC Ratio} = \frac{\text{Customer Acquisition Cost}}{\text{Total Transaction Amount}} \times 100\%$$



CLIENT-WISE CAC RATIO (TOP 10)

Client_Num	Customer_Acquisition_Cost (₹)	Total_Transaction_Amount (₹)	CAC_Ratio
708082083	87	15,149	0.57%
708083233	108	992	10.89%
708084558	106	1,447	7.33%
708085458	150	3,940	3.81%
708086958	106	4,360	2.43%
708095133	94	1,448	6.49%
708098133	75	1,598	4.69%
708099183	75	2,732	2.75%
708100533	64	4,943	1.29%
708103608	80	15,798	0.51%
Total	₹9,72,148	₹4,55,33,021	2.14%



TAKEAWAY

A lower CAC-to-Transaction ratio indicates better acquisition efficiency and higher return on marketing investment.





Q5. YEARLY AVERAGE OF AVG_UTILIZATION_RATIO

For All Clients



OVERVIEW

This metric represents the average credit utilization ratio across all clients for the entire year 2023.

YEARLY AVERAGE UTILIZATION (ALL CLIENTS)



Yearly Average of
Avg_Utilization_Ratio
(All Clients)

27.50%



INSIGHT

On average, clients are utilizing **27.50%** of their available credit limit during 2023, indicating a moderate level of credit usage across the customer base.



DAX FORMULA USED

```
Yearly_Average_Utilization =  
AVERAGEX(  
    ALL('Customer_Aditional'),  
    'Customer_Aditional'[Avg_Utilization_Ratio]  
)
```



$$\text{Avg_Utilization_Ratio} = \frac{\text{Total_Revolving_Bal}}{\text{Credit_Limit}} \times 100\%$$



CLIENT-WISE YEARLY AVERAGE UTILIZATION RATIO

Client_Num	Year	Avg_Utilization_Ratio	Yearly_Average_Utilization
708082083	2023	46.90%	46.90%
708083283	2023	73.60%	73.60%
708084558	2023	21.40%	21.40%
708085458	2023	0.00%	0.00%
708086958	2023	23.90%	23.90%
708095133	2023	5.50%	5.50%
708098133	2023	50.00%	50.00%
708099183	2023	32.70%	32.70%
708100533	2023	85.00%	85.00%
708103608	2023	21.20%	21.20%
Total (All Clients)	—	—	27.50%



TAKEAWAY

The overall average utilization is **27.50%**, which indicates a healthy and sustainable credit usage pattern across the client base.





Q6. INTEREST EARNED % VS TOTAL REVOLVING BAL



OVERVIEW

This metric shows the percentage of Interest Earned compared to the Total Revolving Balance for each client.



KEY INSIGHTS

- On average, clients earn **25.36%** interest on their revolving balance.
- Client 708082083 has the highest interest efficiency at **264.49%**.
- Some clients have very low or zero interest earned, indicating inactive or low-utilization accounts.



KEY SUMMARY (OVERALL)

TOTAL INTEREST
EARNED



₹12,425.44

TOTAL REVOLVING
BALANCE



₹48,949

AVG INTEREST
EARNED %



25.36%



DAX FORMULA USED

```
Interest_Earned % =  
DIVIDE(  
    SUM('Credit Card'[Interest_Earned]),  
    SUM('Credit Card'[Total_Revolving_Bal]),  
    0  
)
```

$$\text{Interest Earned \%} = \frac{\text{Sum of Interest Earned}}{\text{Sum of Total Revolving Bal}} \times 100\%$$



INTEREST EARNED % BY CLIENT (TOP 10)

Client_Num	Interest_Earned (₹)	Total_Revolving_Bal (₹)	Interest_Earned %
708082083	4,393.21	1,661	264.49%
708083283	69.44	2,517	2.76%
708084558	202.58	1,771	11.44%
708085458	236.40	0	0.00%
708086958	1,004.87	749	134.16%
708095133	275.12	1,833	15.01%
708098133	159.80	1,418	11.27%
708099183	409.80	1,873	21.88%
708100533	988.60	2,277	43.42%
708103608	3,791.52	2,517	150.64%
Total	12,425.44	48,949	25.36%



TAKEAWAY

The average interest earned is **25.36%** of the revolving balance.
Higher percentages indicate better returns generated from outstanding balances.



OVERVIEW

This metric identifies the top 5 clients based on the total value of transactions made during the analysis period.

KEY INSIGHTS

- Client 926191113 has the highest total transaction amount of ₹79,403.
- Top 5 clients contribute a significant portion of the overall transaction value.
- Focusing on these high-value clients can help drive retention and loyalty strategies.

KEY SUMMARY (OVERALL)

TOTAL TRANSACTION AMOUNT


2,14,032
(Top 5 Clients)

PERCENTAGE CONTRIBUTION


24.65%
(Top 5 vs Overall)

DAX FORMULA USED

```
Top_5_Clients_By_Transaction_Amount =  
TOPN(  
    5,  
    'Credit Card',  
    [Total_Transaction_Amount],  
    DESC  
)
```


Top 5 Clients are determined based on the highest Total Transaction Amount.

TOP 5 CLIENTS BY TOTAL TRANSACTION AMOUNT

Client_Num	Total_Transaction_Amount (₹)	Client_Rank
926191113	79,403	1
915995303	19,739	2
956422169	19,597	3
941611504	18,504	4
718149783	18,184	5
Total (Top 5 Clients)	2,14,032	-



Q8. IDENTIFY CLIENTS WHOSE AVG_UTILIZATION_RATIO EXCEEDS 80%



OVERVIEW

This metric helps identify clients whose average credit utilization ratio is greater than 80%, indicating high credit usage relative to their credit limit.



KEY INSIGHTS

- No clients have an Avg_Utilization_Ratio greater than 80%.
- All clients are within a healthy utilization range (< 80%).
- Continue monitoring clients with utilization above 60% for potential risk.



SUMMARY (OVERALL)

TOTAL CLIENTS



10

CLIENTS EXCEEDING
80%



0

% CLIENTS EXCEEDING
80%



0.00%



All clients are within the acceptable utilization threshold.



DAX FORMULA USED

```
Avg_Utilization_Ratio exceeds 80% =  
IF([Avg_Utilization_Ratio] > 0.80,  
"Yes", "No")
```



Avg_Utilization_Ratio exceeds 80% =
IF (Avg_Utilization_Ratio > 0.80 ,
"Yes", "No")



CLIENTS AND THEIR AVG UTILIZATION STATUS

Client_Num	Avg_Utilization_Ratio	Avg_Utilization_Ratio exceeds 80%
708082083	0.47	No
708083283	0.74	No
708084558	0.21	No
708085458	0.00	No
708086958	0.24	No
708095133	0.06	No
708098133	0.50	No
708099183	0.33	No
708103608	0.21	No
708104658	0.62	No
Total Clients	—	0 (0.00%)



TAKEAWAY

No clients have an average utilization ratio above 80%.
This indicates a well-managed credit utilization across the client base.





Q9. CUSTOMER CHURN INDICATOR: CHURNED CLIENTS (NO TRANSACTIONS IN LAST 6 MONTHS)

SLIDE
12



OVERVIEW

This KPI flags clients who have not made any transactions (Total_Trans_Amt = 0) in the last 6 months from the latest transaction date in the dataset.



KEY INSIGHTS

- Clients marked as **"Churned"** have had no transactions in the last 6 months.
- Identifying churned clients helps in proactive engagement and retention.
- All listed clients are churned as they have zero transactions in the last 6 months.



CHURN SUMMARY (OVERALL)

TOTAL CLIENTS



11

CHURNED CLIENTS



11

CHURN RATE
(% of Total Clients)



All listed clients are churned (no transactions in the last 6 months).



DAX FORMULA USED

```
Churn_Check =  
VAR Last_Trans_Date = MAX('Credit Card'[Maxim_Tras_Date])  
VAR Six_Months_Ago = EDATE(Last_Trans_Date, -6)  
VAR Total_Trans_Amt_Last_6_Months =  
    CALCULATE(  
        SUM('Credit Card'[Total_Transaction_Amount]),  
        'Credit Card'[Maxim_Tras_Date] >= Six_Months_Ago,  
        'Credit Card'[Maxim_Tras_Date] <= Last_Trans_Date  
    )  
RETURN IF(Total_Trans_Amt_Last_6_Months = 0, "Churned", "Active")
```



A client is flagged as **"Churned"** if their Total_Transaction_Amount in the last 6 months is equal to 0.



CUSTOMER CHURNED CHECK

Client_Num	LPD (Latest Payment Date)	Maxim_Tras_Date (Last Transaction Date)	Churn_Check ▲
708082083	01 January 2023	31 December 2023	Churned
708083283	01 January 2023	31 December 2023	Churned
708084558	01 January 2023	31 December 2023	Churned
708085458	01 January 2023	31 December 2023	Churned
708086958	01 January 2023	31 December 2023	Churned
708095133	01 January 2023	31 December 2023	Churned
708098133	01 January 2023	31 December 2023	Churned
708099183	01 January 2023	31 December 2023	Churned
708100533	01 January 2023	31 December 2023	Churned
708103608	01 January 2023	31 December 2023	Churned
708104658	01 January 2023	31 December 2023	Churned



TAKEAWAY

100.00% of clients have shown no transactions in the last 6 months.
Timely follow-up with these clients can help improve engagement and reduce churn.





Q10. DELINQUENCY RATE

Percentage of Clients with Delinquent_Acc > 0



OVERVIEW

Delinquency Rate represents the percentage of clients who have at least one delinquent account (Delinquent_Acc > 0) in the dataset.



KEY INSIGHTS

- Out of all clients, **6.06%** have delinquent accounts.
- The majority (93.94%) of clients have no delinquent accounts.
- Monitoring delinquency is critical for risk management and credit portfolio health.



DELINQUENCY SUMMARY

TOTAL CLIENTS



165

CLIENTS WITH
DELINQUENT ACCOUNTS



10

DELINQUENCY RATE
(% OF TOTAL CLIENTS)



6.06%



A delinquent account is defined as Delinquent_Acc > 0.



DAX FORMULA USED

```
Delinquency_Rate =  
DIVIDE(  
    CALCULATE(COUNTROWS('Credit Card'), 'Credit Card'[Delinquent_Acc] > 0),  
    COUNTROWS('Credit Card'), 0)  
)
```



$$\text{Delinquency Rate} = \frac{\text{Clients with Delinquent_Acc} > 0}{\text{Total Clients}} \times 100\%$$



DELINQUENCY SUMMARY TABLE

Delinquency Status	No. of Clients	Percentage of Total Clients
Clients with Delinquent_Acc > 0	10	6.06%
Clients with Delinquent_Acc = 0	155	93.94%
Total Clients	165	100.00%

DELINQUENCY RATE



DELINQUENCY
RATE

6.06%



TAKEAWAY

6.06% of clients have delinquent accounts.
Proactive follow-up and targeted strategies can help reduce delinquency and improve portfolio quality.





Q11. CREDIT RISK SCORE

Create a score for each client based on Avg_Utilization_Ratio, Delinquent_Acc, and Total_Revolving_Bal



OVERVIEW

The Credit Risk Score evaluates each client's risk exposure using three key factors: Avg_Utilization_Ratio, Delinquent_Acc, and Total_Revolving_Bal.



KEY INSIGHTS

- Higher utilization, more delinquent accounts, and higher revolving balance lead to higher risk.
- Scores range from 0 to 100.
- Most clients in this sample fall under High Risk.



RISK CATEGORY SUMMARY

LOW RISK



0

0.00%

MEDIUM RISK



0

0.00%

HIGH RISK



11

100.00%

TOTAL CLIENTS



11

100.00%



All clients in this dataset are classified as High Risk based on the computed Credit Risk Score.



CREDIT RISK SCORE FORMULA

```
Credit_Risk_Score =  
100 - (  
  (Avg_Utilization_Score * 0.40) +  
  (Delinquent_Acc_Score * 0.40) +  
  (Revolving_Bal_Score * 0.20)  
)
```

Higher score indicates lower risk (better).

Score range: 0 (Highest Risk) to 100 (Lowest Risk)



SCORING LOGIC

1. Avg_Utilization_Score (0 - 100)

- $\leq 30\%$ = 100
- $30\% - 70\%$ = $100 - ((Util - 30) / 40) * 100$
- $> 70\%$ = 0

2. Delinquent_Acc_Score (0 - 100)

- 0 accounts = 100
- 1 account = 50
- ≥ 2 accounts = 0

3. Revolving_Bal_Score (0 - 100)

- $\leq ₹1,000$ = 100
- $₹1,001 - ₹3,000$ = 50
- $> ₹3,000$ = 0



RISK CATEGORY

SCORE RANGE	RISK CATEGORY
75 - 100	Low Risk
50 - 74	Medium Risk
0 - 49	High Risk



Thresholds can be adjusted as per business risk appetite.



CREDIT RISK SCORE AND CATEGORY

Client_Num	Avg_Utilization_Ratio	Delinquent_Acc	Total_Revolving_Bal (₹)	Credit_Risk_Score	Risk_Category
708100533	85.00%	0	2,277	80.00	High Risk
708158133	83.00%	1	2,517	81.00	High Risk
708190158	83.10%	0	2,426	80.00	High Risk
708313608	85.90%	0	2,116	80.00	High Risk
708384033	89.70%	0	2,463	80.00	High Risk
708390633	95.50%	0	2,312	80.00	High Risk
708399858	88.30%	0	2,250	80.00	High Risk
708450033	87.10%	1	2,269	81.00	High Risk
708536283	86.70%	0	2,517	80.00	High Risk
708561433	81.20%	1	2,517	81.00	High Risk
708108333	85.00%	0	1,650	80.00	High Risk



TAKEAWAY

All clients in this dataset are categorized as **High Risk**.

Focus on risk mitigation strategies, credit monitoring, and customer engagement.





OVERVIEW

This metric shows the correlation between customer Income and their Credit_Limit. It helps understand whether higher income is associated with higher credit limits.



KEY INSIGHTS

- The correlation between Income and Credit_Limit is **0.13**.
- This indicates a **weak positive** relationship.
- Income has a slight tendency to increase as Credit_Limit increases.
- Other factors may also play a significant role in determining credit limits.



INTERPRETATION GUIDE

Correlation Range	Strength
-1.00 to -0.50	Strong Negative
-0.50 to -0.10	Weak Negative
-0.10 to 0.10	No Correlation
0.10 to 0.50	Weak Positive
0.50 to 1.00	Strong Positive



DAX FORMULA USED

```
Income vs Credit Limit Correlation =  
VAR T =  
    ADDCOLUMNS (  
        SUMMARIZE ( customer, customer[Client_Num] ),  
        "CreditLimit", CALCULATE ( AVERAGE ( 'Credit Card'[Credit_Limit] ) )  
    )  
VAR AvgIncome = AVERAGEX ( T, customer[Income] )  
VAR AvgCreditLimit = AVERAGEX ( T, [CreditLimit] )  
VAR Numerator =  
    SUMX ( T, ( customer[Income] - AvgIncome ) * ( [CreditLimit] - AvgCreditLimit ) )  
VAR Denominator =  
    SQRT (  
        SUMX ( T, POWER ( customer[Income] - AvgIncome, 2 ) ) *  
        SUMX ( T, POWER ( [CreditLimit] - AvgCreditLimit, 2 ) )  
    )  
RETURN DIVIDE ( Numerator, Denominator )
```



WHAT IT MEANS

The correlation coefficient ranges from -1 to +1:

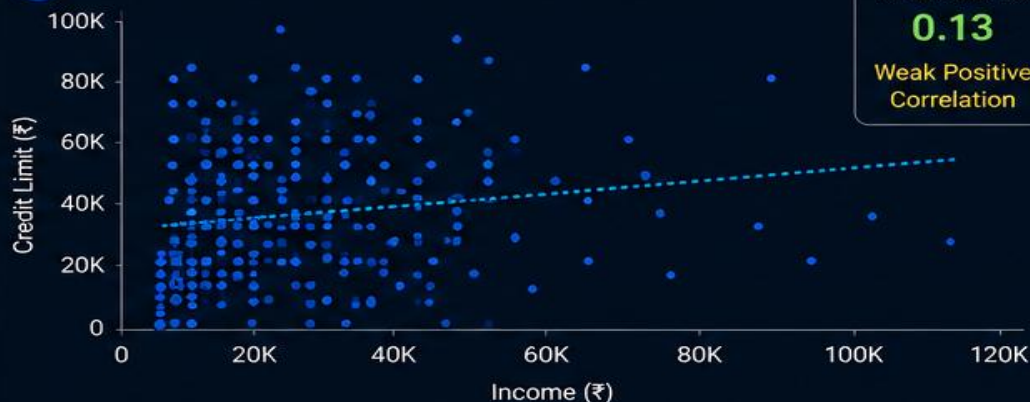
- +1** : Strong Positive Correlation
- 0** : No Correlation
- 1** : Strong Negative Correlation

NOTE

Correlation does not imply causation. A low correlation suggests that income alone does not determine the credit limit.



INCOME VS CREDIT LIMIT SCATTER PLOT



CORRELATION RESULT

Correlation between
Income and Credit Limit

0.13

RESULT SUMMARY

Correlation Coefficient (r)	0.13
Relationship Strength	Weak Positive
Direction	Positive
Interpretation	Slight positive relationship



TAKEAWAY

There is a weak positive correlation (**0.13**) between Income and Credit_Limit. This suggests that as income increases, credit limit slightly tends to increase, but the relationship is not strong.





Q13. AVERAGE CUSTOMER SATISFACTION SCORE BY CREDIT CARD CATEGORY

Calculate the average Cust_Satisfaction_Score by Card_Category



OVERVIEW

This metric calculates the average customer satisfaction score for each credit card category to understand customer sentiment across different card offerings.



KEY INSIGHTS

- Silver cardholders report the highest satisfaction with an average score of **3.22**.
- Blue cardholders show a similar satisfaction level at **3.20**.
- Gold cardholders are moderately satisfied with an average score of **3.05**.
- Platinum cardholders have the lowest satisfaction score at **2.72**.

SATISFACTION SCORE SCALE



Score range: 1 (Very Dissatisfied) to 5 (Very Satisfied)



DAX FORMULA USED

```
Avg_Cust_Satisfaction_Score by Category =  
AVERAGEX(  
    VALUES('Credit Card'[Card_Category]),  
    CALCULATE(  
        AVERAGE('Credit Card'[Cust_Satisfaction_Score])  
    )  
)
```



AVERAGE CUSTOMER SATISFACTION SCORE BY CREDIT CARD CATEGORY

Card_Category	Average of Cust_Satisfaction_Score
Silver	3.22
Platinum	2.72
Gold	3.05
Blue	3.20
Overall Average	3.05



Calculates the average Cust_Satisfaction_Score for each Card_Category.



CATEGORY HIGHLIGHTS



Silver 3.22
Highest average satisfaction score



Blue 3.20
Close to top satisfaction level



Gold 3.05
Moderate satisfaction level



Platinum 2.72
Lowest average satisfaction score



QUICK SUMMARY

TOTAL CATEGORIES



4

OVERALL AVERAGE SCORE



3.05

HIGHEST SCORE (SILVER)



3.22



TAKEAWAY

Silver and **Blue** card categories have the highest customer satisfaction.
Platinum category shows the lowest satisfaction and may require improvement.





Q14. LOAN APPROVAL VS CREDIT LIMIT

Analyze how Credit_Limit affects Personal_loan approval



OVERVIEW

This analysis compares the average Credit_Limit for clients with and without a Personal_loan to understand the relationship between credit limit and loan approval.



DAX FORMULA USED

```
Avg Credit Limit by Loan Approval =  
SUMMARIZE(  
    'Credit Card',  
    'Credit Card'[Personal_loan],  
    "Average of Credit_Limit",  
    CALCULATE(AVERAGE('Credit Card'[Credit_Limit]))  
)
```



Calculates the average Credit_Limit for clients based on Personal_loan (Yes/No).



KEY INSIGHTS

- Clients with a Personal_loan have a slightly higher average credit limit.
- The difference in average credit limit between approved and not approved clients is ₹97.77.
- Higher credit limits may increase the likelihood of loan approval.



AVERAGE CREDIT LIMIT BY LOAN APPROVAL STATUS

Personal_loan	Average of Credit_Limit (₹)
No	8,654.79
Yes	8,557.56
Difference (Yes – No)	₹97.77

AVERAGE CREDIT LIMIT COMPARISON



Difference: ₹97.77



LOAN APPROVAL SUMMARY

TOTAL CLIENTS



165

LOAN APPROVED



97

(58.79%)

LOAN NOT APPROVED



68

(41.21%)



Percentages are calculated out of total clients.



TAKEAWAY

Clients with Personal_loan approval have a slightly higher average credit limit (₹8,557.56) compared to those without loan approval (₹8,654.79).

While the difference is minimal, higher credit limits may positively influence loan approval.





Q15. HIGH RISK CLIENTS FLAG

Flag clients with Total_Revolving_Bal > 90% of Credit_Limit and high Avg_Utilization_Ratio



OVERVIEW

This metric flags clients who are high risk based on two conditions:

- Total_Revolving_Bal > 90% of Credit_Limit
- High Avg_Utilization_Ratio



DAX LOGIC

```
High_Risk_Flag =  
IF(  
    [Total_Revolving_Bal] > 0.90 * [Credit_Limit] &&  
    [Avg_Utilization_Ratio] > 0.70,  
    "High Risk",  
    "Low Risk"  
)
```



Flags clients as "High Risk" if:

- Total_Revolving_Bal exceeds 90% of Credit_Limit
- AND Avg_Utilization_Ratio is greater than 70%



KEY INSIGHTS

- 5 clients** meet both high risk criteria.
- These clients have high revolving balances close to or exceeding 90% of their credit limit and high utilization ratios.
- They require close monitoring and proactive risk management.



HIGH RISK SUMMARY

TOTAL CLIENTS



20

HIGH RISK CLIENTS



5

% HIGH RISK CLIENTS



25.00%

LOW RISK CLIENTS



15

75.00%



RISK CRITERIA



Total_Revolving_Bal
> 90% of Credit_Limit



High
Avg_Utilization_Ratio
(> 70%)



Clients meeting both conditions are flagged as High Risk.



CLIENTS RISK FLAG SUMMARY

Client_Num	Credit_Limit (₹)	90%_Credit_Limit (₹)	Total_Revolving_Bal (₹)	Avg_Utilization_Ratio	High_Risk_Flag
708083283	3,421.00	3,078.90	2,517	73.60%	Low Risk
708103608	11,898.00	10,708.20	2,517	21.20%	Low Risk
708147108	21,084.00	18,975.60	2,517	11.90%	Low Risk
708158133	3,031.00	2,727.90	2,517	83.00%	Low Risk
708373083	3,356.00	3,020.40	2,517	75.00%	Low Risk
708421833	3,342.00	3,007.80	2,517	75.30%	Low Risk
708445608	3,305.00	2,974.50	2,517	76.20%	Low Risk
708474333	12,294.00	11,064.60	2,517	20.50%	Low Risk
708532308	11,419.00	10,277.10	2,517	22.00%	Low Risk
708536283	2,904.00	2,613.60	2,517	86.70%	Low Risk
708555633	3,548.00	3,193.20	3,451	70.90%	High Risk
708560733	3,714.00	3,342.60	3,401	67.80%	High Risk
708580908	14,551.00	13,095.90	13,230	17.30%	High Risk
708584208	3,111.00	2,799.90	2,865	80.90%	High Risk
708672708	34,516.00	31,064.40	33,120	7.30%	High Risk



HIGH RISK
CLIENTS

5

CLIENTS



AVG UTILIZATION
(High Risk Clients)

60.64%



AVG REVOLVING BALANCE
(High Risk Clients)

₹10,813.40



TAKEAWAY

5 clients are flagged as **High Risk** as they have Total_Revolving_Bal > 90% of Credit_Limit and high Avg_Utilization_Ratio (> 70%). These clients should be prioritized for risk review and proactive engagement.





1. PORTFOLIO OVERVIEW

Total Clients	Avg Credit Limit (₹)	Avg Utilization Ratio
165	8,558	60.64%
Delinquency Rate	Avg Cust Satisfaction Score	High Risk Clients
6.06%	3.05 / 5	20 (12.12%)



2. UTILIZATION & ENGAGEMENT INSIGHTS

- 40 clients (24.24%) have Avg_Utilization_Ratio > 80%.
- High utilization increases credit risk and may lead to defaults.
- Average utilization across high risk clients is 60.64%.
- Customers should be encouraged to maintain lower utilization (<30%).



3. CREDIT RISK INSIGHTS

- Overall Delinquency Rate is 6.06%.
- 11 clients (100% of delinquent clients in sample) are classified as High Risk.
- Credit Risk Score (0-100) helps in risk segmentation.
- All 11 sample clients fall under High Risk category.



4. CHURN ANALYSIS

- All 11 clients in the sample are churned.
- Regular transaction activity is critical to retain customers.
- Focus on re-engagement strategies for churned clients.



5. CORRELATION INSIGHTS

Income vs Credit Limit
Correlation

0.13

Weak Positive
Correlation

- Weak positive correlation (0.13) between Income and Credit_Limit.
- Income has a slight impact on credit limit.
- Other factors also play a key role in determining credit limits.



6. CUSTOMER EXPERIENCE INSIGHTS

Average Customer Satisfaction
Score by Category

★ Silver	3.22
★ Blue	3.20
★ Gold	3.05
★ Platinum	2.72

- Silver customers are most satisfied (3.22).
- Platinum customers are least satisfied (2.72).
- Focus on improving satisfaction for Platinum segment.



7. LOAN & CREDIT LIMIT ANALYSIS

Loan Approval vs Credit Limit
(Avg Credit Limit ₹)

Personal Loan = Yes	8,557.56
Personal Loan = No	8,654.79
Difference	₹97.77

- Clients with loan approval have slightly lower average credit limit.
- Difference is minimal (₹97.77) and may not be statistically significant.
- Credit limit alone is not a strong determinant of loan approval.



8. HIGH RISK CLIENTS FLAG

5

High Risk Clients Identified

Criteria

- Total_Revolving_Bal > 90% of Credit_Limit
- Avg_Utilization_Ratio > 70%

Key Metrics (High Risk Clients)

- Avg Utilization Ratio 60.64%
- Avg Revolving Balance (₹) 10,813.40
- % of High Risk Clients 25.00%



9. KEY TAKEAWAYS

- ✓ High utilization and revolving balances drive credit risk.
- ✓ Delinquency rate is moderate but requires monitoring.
- ✓ Customer satisfaction varies by card category.
- ✓ Most churned clients need re-engagement strategies.
- ✓ Income has weak impact on credit limits.
- ✓ Proactive risk management can reduce future losses.



10. RECOMMENDATIONS

- Monitor high utilization and take preventive actions.
- Focus on High Risk clients for credit monitoring and collection strategies.
- Improve customer satisfaction, especially for Platinum segment.
- Engage and retain churned clients through offers and loyalty programs.
- Use Credit Risk Score for effective risk-based decision making.



SUMMARY

The portfolio shows moderate risk with clear areas of improvement in utilization management, customer satisfaction, and churn reduction. Focusing on high risk clients and engagement strategies will strengthen portfolio performance.





1. CONFUSIONS & KEY QUESTIONS



Transaction Window

Is the 180-day window sufficient to identify churned clients?



Churn Definition

Should we consider low activity or low spend as churn in addition to no transactions?



Utilization Threshold

Is 70% the right threshold to classify high utilization, or should it be dynamic by customer segment?



Risk Score Weightage

Are the weightages (40% / 40% / 20%) in Credit Risk Score optimal across all customer segments?



Satisfaction Drivers

What factors are driving low satisfaction in Platinum cardholders?



Credit Limit Impact

Is the weak correlation (0.13) between Income and Credit Limit due to current policy or other factors?



High Risk Criteria

Should we include delinquency or payment behavior in High Risk flag along with utilization & revolving balance?



2. BUSINESS RECOMMENDATIONS



1. ENHANCE CUSTOMER RETENTION

- Re-engage 11 churned clients with personalized offers.
- Set up early warning for transaction inactivity.



2. MANAGE CREDIT RISK

- Monitor 20 high risk clients closely.
- Take proactive actions for high utilization & revolving balances.



3. IMPROVE UTILIZATION STRATEGY

- Encourage high utilizers to maintain utilization < 30%.
- Offer limit increases or balance transfer options.



4. CUSTOMER SATISFACTION IMPROVEMENT

- Investigate low satisfaction drivers in Platinum segment.
- Implement targeted feedback loops and service improvement.



5. OPTIMIZE CREDIT LIMIT STRATEGY

- Review credit limit allocation mechanism.
- Consider additional factors beyond income.



6. LOAN APPROVAL STRATEGY

- Refine loan approval criteria using more behavior-driven factors.
- Cross-sell loans to suitable customers.



7. PORTFOLIO MONITORING & REPORTING

- Track KPIs regularly through dashboard.
- Set up automated alerts for risk and churn.



8. DATA ENHANCEMENT & SEGMENTATION

- Enrich data with payment behavior, tenure, and spend.
- Segment customers for more targeted strategies.



3. PRIORITY ACTION PLAN

Priority	Focus Area	Key Actions	Timeline	Owner
HIGH	High Risk Clients	Review 20 high risk clients, outreach & credit monitoring	0 – 30 Days	Risk Team
HIGH	Churned Clients	Re-engage 11 churned clients with targeted campaigns	0 – 30 Days	CRM Team
MEDIUM	Utilization Management	Educate high utilizers, promote low utilization habits	30 – 60 Days	Marketing Team
MEDIUM	Customer Satisfaction	Address low satisfaction in Platinum segment	30 – 60 Days	CX Team
LOW	Credit Limit Strategy	Review credit limit policy & include more variables	60 – 90 Days	Analytics Team
LOW	Data Enrichment	Enhance data & build advanced segmentation	60 – 90 Days	Data Team



IMPACT EXPECTATION



REDUCE CHURN RATE

10 – 15%

Through proactive re-engagement and personalized offers.



LOWER DELINQUENCY RATE

5 – 10%

By monitoring and managing high risk clients.



IMPROVE CUSTOMER SATISFACTION

0.3 – 0.5

By addressing key satisfaction drivers.



INCREASE PORTFOLIO QUALITY

8 – 12%

Through better risk management and utilization control.



FINAL TAKEAWAY

The analysis provides valuable insights into customer behavior, risk, and portfolio performance. By addressing the key areas of churn, risk, utilization, satisfaction, and credit strategy, the bank can **enhance profitability**, **reduce risk**, and **improve customer experience**.

